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Multi-Agent Reinforcement Learning with Structured Communication Optimization in CityLearn

Midterm Report

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Chapter 1

Introduction

Many community buildings are no longer just passive electricity users. Modern homes can have solar panels, batteries, HVAC equipment for heating and cooling, and other controllable devices that support local energy flexibility. If homes in a community are grouped and controlled together, their total electricity use can be moved away from peak periods and made closer to a desired target load profile. The challenge is that each home only sees its own situation, but the success of the controller is judged by the behaviour of the whole building portfolio.

This midterm report studies portfolio load tracking in the CityLearn simulation environment. CityLearn is an open simulation environment for building energy control and for comparing reinforcement learning algorithms [20, 19, 14]. The project uses the IEA EBC Annex 96 Common Exercise 1 (CE1) as a benchmark setting. In this task, 25 single-family homes need to follow a daily reference load profile as a group. The controller can adjust battery operation and HVAC-related flexibility for both heating and cooling. The evaluation considers not only load tracking error, but also indoor comfort and secondary metrics such as total energy use, peak demand, peak-to-valley ratio, load factor, ramping, and fairness. Cost and carbon emissions are part of the Annex 96 metric set, but the CE1 schemas used in this project do not provide pricing or carbon-intensity files, so these two fields are not interpreted in the current experiments. In this report, the reward function is designed around the three primary objectives, while the secondary metrics are used only for evaluation rather than direct optimization.

The core research question is:

In the CityLearn portfolio load tracking task, what kinds of MARL structures and communication designs are more effective, and why are they effective?

This project is organized around three comparisons. First, it compares rule-based control and independent building control with MARL-based coordination to test whether learned coordination is actually useful for CE1. Second, using the stable on-policy method MAPPO as the current example, it compares different MARL structures, including grouped and ungrouped policies, communication and no-communication variants, and intra-group and inter-group communication. Third, it will compare different types of MARL, such as MAPPO, MASAC, or value-based methods, to understand which learning style is more suitable for this task.

At the current stage, MAPPO is used as the main experimental backbone. This is a practical choice rather than a final claim that MAPPO is the best algorithm. MAPPO is an on-policy centralized-training, decentralized-execution method, and PPO-style updates are usually stable and easier to debug [16, 22]. This is useful for first building a reliable CityLearn pipeline with grouped actors, a shared critic, and communication modules. After this pipeline is stable, the project can compare MAPPO with other directions such as SAC/MASAC or DQN-style value-based methods.

The communication part is therefore an optimization and innovation on top of the current MAPPO backbone. It studies whether agents should share information, and if so, whether simple mean-message communication, PowerNet-style structured information passing, graph-attention communication, attention-based communication, or weighted same-group/other-group communication is more effective.

The current stage of the project has produced a working research codebase and a clear experimental plan. The completed work includes:

- An RBC notebook baseline for non-learning comparison in Texas and Vermont;
- Implementation of CityLearn wrappers for independent PPO/SAC and MAPPO training;
- Implementation of standard MAPPO and grouped MAPPO baselines;
- Implementation of grouped actor policies using K-means clustering over building flexibility features;

- Implementation of several differentiable communication modules, including CommNet-style mean pooling, PowerNet-style structured information passing, GAT-style graph attention, TarMAC-style attention, DIAL-inspired global communication, and weighted same-group/other-group communication;

- Implementation of logging, checkpointing, testing, and Annex 96 metric export utilities.

The project has a completed Vermont benchmark comparison for the selected controllers, together with exported test-month metrics, ranking tables, and load-tracking figures generated from the `results`, `wandb`, and `target_folder` outputs. The current Vermont results already support a meaningful comparison of load tracking, comfort, peak demand, fairness, and communication design. At present, `mappo_grouped_powernet_global` is the strongest overall method, while the default weighted communication setting leads the primary-only ranking. Texas RL experiments and multi-seed validation remain part of the next stage, although the Texas training pipeline has already been implemented and can be launched directly. Vermont was prioritized first because its district target is less flat and is therefore more useful for observing performance differences during the early training stage.

Chapter 2 reviews the relevant literature and explains where the project fits. Chapter 3 describes the implemented method and current progress. Chapter 4 gives the completed Vermont evaluation and the main result analysis. Chapter 5 summarizes the midterm outcomes and next steps.

Chapter 2

Literature Review

2.1 Building Energy Flexibility and CityLearn

Reinforcement learning for building energy management has moved from isolated HVAC control toward coordinated control of flexible communities. The motivation is clear: buildings consume a large share of electricity and can provide demand flexibility, but controllers must balance energy, comfort, cost, carbon, and operational constraints. Recent surveys emphasize that RL is promising for adaptive building control, while also noting practical barriers such as safety, sample efficiency, sim-to-real transfer, and benchmark comparability [23, 21].

CityLearn helps address the comparability issue. CityLearn v1 introduced an OpenAI Gym-style environment for demand response using deep RL [20]. The later CityLearn work standardized multi-agent RL research for urban demand response and made comparison between algorithms more reproducible [19]. CityLearn v2 extends the environment with energy-flexible, resilient, occupant-centric, and carbon-aware community control, including richer distributed energy resource models and updated evaluation use cases [14]. This makes CityLearn an appropriate environment for this project because the research question is not simply whether an agent can reduce energy cost, but whether multiple building agents can coordinate under shared portfolio-level objectives.

2.2 Annex 96 CE1 Experimental Setting

This project uses Annex 96 Common Exercise 1 (CE1) as a benchmark setting in CityLearn. Each CE1 schema defines 25 single-family homes, one weather file, one district target file, and one energy-simulation file per home. The district target file provides the reference load that the whole portfolio should follow, while the per-building files provide local demand, indoor temperature, solar generation, and other time-series observations. The important point for this project is that CE1 is a portfolio load-tracking task, not a physical network control task.

The CE1 schemas used here do not provide a feeder map, floor-plan distribution, geographic coordinates, or an adjacency list between homes. Therefore the current grouping method cannot use physical proximity or electrical topology. Instead, buildings are grouped using flexibility-related device features that are available after CityLearn initializes the environment, mainly electrical storage capacity and total HVAC nominal power. This matches the K-means implementation used in the experiments.

Another data limitation is pricing and carbon intensity. The CE1 schema contains observation names for electricity pricing and carbon intensity, but the VT and TX CE1 building entries used in this project set both the pricing file and the carbon-intensity file to null. CityLearn then fills these series with zeros. As a result, cost and carbon-emission metrics still have calculation formulas in CityLearn, but they are zero or null in the current experiments and should not be interpreted as meaningful performance outcomes.

2.3 MARL for Energy Systems

Multi-agent reinforcement learning is a natural formulation for energy communities because each building, inverter, or household can be treated as an agent. General MARL surveys highlight three recurring difficulties: non-stationarity from concurrently learning agents, partial observability, and credit assignment under shared rewards [25]. In energy systems these issues are amplified by heterogeneous devices, different comfort states, network constraints, and long temporal dependencies.

Recent energy-system studies show why MARL is appropriate but also why structure is necessary. PowerGridworld provides a Gym-style environment for multi-agent power-system control and demonstrates MARL with PPO and MADDPG [1]. Charbonnier et al. study scalable residential energy flexibility and show that centralized-but-factored critics can improve coordination and scalability for decentralized homes [2]. Physics-informed MARL for distribution-network voltage control further shows that graph and physical structure can improve robustness and sample efficiency [24]. These works support the design principle used in this project: scalable energy MARL should not treat agents as an unstructured list; it should use building similarity, graph structure, or communication structure.

2.4 Rule-Based and Single-Agent Baselines

Rule-based control (RBC) is an important non-learning baseline in CityLearn. An RBC controller uses fixed rules, often based on hour-of-day schedules, to charge or discharge storage and to set heating or cooling device availability. It is simple, transparent, and cheap to run, but it cannot adapt its policy to the CE1 reference profile or to interactions between buildings. In this project, RBC is therefore used as a practical reference point: a MARL method should eventually do more than repeat a fixed schedule.

Independent single-agent RL baselines are also important. PPO is a robust on-policy policy-gradient method that limits destructive policy updates through a clipped surrogate objective [16]. It is often used as a stable baseline because it is comparatively simple and reliable. SAC is an off-policy maximum-entropy actor-critic method that learns stochastic policies and encourages exploration through entropy regularization [8, 9]. In this project, independent PPO and SAC provide baseline controllers that answer a basic question: how much performance can be obtained without explicit inter-building coordination?

On-policy and off-policy methods also differ in how they use data. On-policy methods such as PPO and MAPPO collect trajectories using the current policy and then update from those fresh trajectories. This is usually stable and easier to control, but it uses data less efficiently. Off-policy methods such as SAC can store past experience in a replay buffer and reuse it for many updates. This can be more sample-efficient and exploratory, but it can also be more sensitive to distribution shift between old data and the current policy.

However, independent baselines are expected to struggle with portfolio-level tracking. Each building receives only local observations and cannot know how other buildings will respond at the same time step. This motivates moving from independent learning to CTDE and communication-aware MARL.

2.5 Centralized Training with Decentralized Execution

CTDE is the dominant framework for cooperative MARL. MADDPG uses centralized critics that condition on other agents' actions to reduce non-stationarity in actor-critic learning [11]. COMA addresses credit assignment by using a centralized critic with a counterfactual baseline that estimates each agent's marginal contribution [6]. Value-decomposition approaches such as VDN and QMIX decompose a team value function into per-agent components. QMIX adds a monotonic mixing constraint so that decentralized greedy actions remain consistent with the centralized value function [18, 15].

MAPPO is a policy-gradient CTDE method that adapts PPO to cooperative multi-agent settings. Yu et al. show that, with careful implementation, PPO-style on-policy learning is surprisingly effective in cooperative multi-agent games [22]. MAPPO is a good match for CityLearn because CityLearn has a continuous bounded action space, and the portfolio objective is cooperative. This project therefore uses MAPPO as the main learning backbone and reuses the existing on-policy implementation rather than reimplementing the optimizer.

2.6 Parameter Sharing, Grouping, and K-Means

Parameter sharing is a common technique for scaling MARL: multiple agents use the same actor network, reducing the number of trainable parameters and improving sample efficiency. The limitation is that full parameter sharing can be harmful when agents are heterogeneous. Selective parameter sharing addresses this by partitioning agents according to abilities or goals, then sharing parameters within each partition [4]. This idea is directly relevant to the CityLearn portfolio, where buildings differ in battery capacity, thermal flexibility, and load profile.

The grouping method used in this project is K-means clustering. K-means partitions data into K sets by minimizing within-cluster squared distances to cluster centroids [13]. In this implementation, buildings are clustered using flexibility-related features such as battery storage capacity and aggregate HVAC capacity. The result is a grouped actor architecture: agents within the same cluster share an actor, while the full portfolio shares a centralized critic. This provides a concrete link between classical clustering, selective parameter sharing, and building-energy heterogeneity.

2.7 Communication in MARL

Communication is another way to overcome partial observability and non-stationarity. A recent survey of multi-agent deep RL with communication argues that communication broadens agents' effective observations and supports cooperation, but also introduces design choices about message content, receiver selection, communication frequency, and overhead [26]. This is especially relevant for building portfolios: all-to-all communication may be unnecessary or noisy, while no communication may be too weak for aggregate tracking.

Several communication methods guide this project. DIAL trains differentiable communication channels under centralized learning and decentralized execution [7]. CommNet performs differentiable message passing by averaging hidden states from other agents and learning a neural update [17]. TarMAC replaces uniform broadcasting with attention-based targeted communication so that agents can learn whom to attend to [5]. PowerNet, developed for scalable power-grid control, uses local-neighbor communication and is closely aligned with the idea that energy systems have useful graph structure [3].

The communication modules in this project combine these ideas with the CityLearn portfolio structure. CommNet tests simple mean-pooling communication. PowerNet-style modules test more structured information passing: each agent keeps its own encoded feature, aggregates selected peer information, concatenates the two streams, and applies a residual update so the original local information is not overwritten. TarMAC tests learned attention. The weighted same-group/other-group module is a project-specific middle ground: it gives higher weight to messages from similar buildings and lower weight to global context. This method is intended to preserve scalability while still allowing the policy to coordinate at portfolio level.

2.8 Research Gap

The literature shows that RL for building control is active, CityLearn provides a strong benchmark, CTDE methods are effective for cooperative MARL, and communication can improve coordination. The gap addressed here is the combination of these ideas for CityLearn portfolio load tracking:

- Most CityLearn controllers do not explicitly compare multiple structured communication protocols;
- Many MARL communication studies use abstract games rather than building-energy metrics such as comfort, ramping, peak demand, and load tracking. In the current CE1 data, cost and carbon are not interpreted because the pricing and carbon-intensity files are absent;
- Full parameter sharing ignores heterogeneity, while fully separate actors lose sample efficiency;
- Final performance must be judged by portfolio-level tracking and secondary energy impacts, not by reward alone.

2.9 Methodological Contribution

The main methodological contribution of this report is the integration of structured communication into a grouped MARL pipeline for CityLearn portfolio load tracking. More specifically, the project combines K-means-based actor grouping with several communication designs, including mean aggregation, weighted intra-group/inter-group messaging, PowerNet-style structured message passing, graph attention, and TarMAC-style attention. This makes it possible to study not only whether communication helps, but also which communication structures are more suitable for CityLearn portfolio load tracking and why.

This project therefore focuses on grouped MAPPO with structured communication, using CityLearn and Annex 96 metrics as the application-grounded evaluation setting.

Chapter 3

Method and Progress

This chapter describes what has been implemented and what has been completed by the midterm stage. The main training pipeline now runs end-to-end, the selected Vermont benchmark experiments have completed their planned training budgets, and the evaluation utilities export unified test-month metrics, per-building temperature tracking figures, ranking tables, and load-tracking figures. The project can now compare several baselines and communication variants within the same MAPPO framework using the completed Vermont result set.

3.1 Problem Definition

The CityLearn Annex 96 task is treated as a cooperative control problem with $N = 25$ building agents. At time step t, building i observes its own local state o_i^t and chooses a continuous action a_i^t for its controllable devices. The actions of all buildings together affect battery states, indoor conditions, and the total portfolio load. The goal is to make the total load follow a daily reference profile r_t without creating too many comfort violations or other negative side effects.

The project objective is not only to get a high training reward. In practice, the controller should reduce the testing-month tracking error, measured by NMBE and CV-RMSE, while keeping comfort, ramping, peak demand, and fairness in a reasonable range. Cost and carbon emissions are part of the Annex 96 secondary metric set, but the CE1 schemas used here do not provide pricing or carbon-intensity time series, so those two metrics are not interpreted in the current experiments. This is why the method uses a centralized critic and structured communication between actors.

3.2 Reward Function

The current training reward uses the custom `annex96_rewards.ce1.CE1ThreeMetricReward`. It is designed as a trainable surrogate of the three CE1 primary objectives: portfolio load bias, portfolio tracking error, and per-building thermal comfort. For building i at each time step, the reward can be written as

$$r_i = -\Bigg[\frac{w_{\text{nmbe}}|\text{NMBE}| + w_{\text{cv\_rmse}}\text{CV-RMSE}}{N} + w_{\text{comfort}}\Big(w_{\text{binary}}\,\mathbb{I}_{\text{exceed},i} + w_{\text{degree}}\,d_{\text{exceed},i}\Big)\Bigg]. \tag{3.1}$$

Here, the NMBE and CV-RMSE terms in Equation (3.1) are running portfolio-level surrogates shared equally across all buildings, while the comfort term is computed separately for each building. The parameter values used in the current code are given in Equation (3.2):

$$w_{\text{nmbe}} = 1.0, \qquad w_{\text{cv\_rmse}} = 1.0, \qquad w_{\text{comfort}} = 0.8, \qquad w_{\text{binary}} = 1.3, \qquad w_{\text{degree}} = 0.3. \tag{3.2}$$

The binary comfort term adds a penalty whenever the indoor temperature goes outside the seasonal comfort band, while the degree term adds an extra penalty proportional to how far the temperature is outside that band. For the current Vermont experiments, which use the January/February setting, the comfort band is 20°C to 24°C.

3.3 Environment and Baseline Pipeline

The first completed component is the environment interface. CityLearn is wrapped for several control and learning pipelines:

- RBC notebook baseline for non-learning comparison;
- Independent PPO and independent SAC baselines, where each building learns without communication;

- Standard MAPPO, where all buildings use a shared policy and a centralized critic;
- Grouped MAPPO, where clusters of buildings share group-specific actors;
- Communication-aware grouped MAPPO, where a communication layer is inserted between actor encoders and action heads.

For the current standard, grouped, and communication-aware MAPPO experiments, the `mappo_standard/env.py` wrapper returns local observations, centralized observations, continuous action spaces, rewards, done masks, and episode lengths in the tensor format expected by `on-policy-main`. The training scripts then reuse `R_MAPPO`, `R_MAPPOPolicy`, and `SharedReplayBuffer` from that implementation, instead of reimplementing the PPO optimizer and buffer logic. This reduces implementation risk and keeps the research focus on CityLearn adaptation, grouping, and communication.

3.4 Grouped Actor Architecture

The grouped actor design sits between two simple choices. If all buildings share one actor, training is efficient, but the policy may ignore important differences between buildings. If every building has its own actor, the policy can be more flexible, but it needs more data and coordination becomes harder. The grouped design shares an actor only within similar buildings.

Buildings are grouped with K-means using features related to flexibility, mainly battery energy storage capacity and total HVAC capacity. After clustering, buildings in group G_k share the same actor network. All groups still share one centralized critic. The critic sees the full portfolio state, while each actor uses its local building observations and, when enabled, communication-enhanced features.

This architecture has been implemented in the `mappo_grouped` package. The same grouping information is reused by communication-aware variants so that communication effects can be compared against a consistent grouped baseline.

3.5 MAPPO Learning Structure

The learning structure follows centralized training and decentralized execution. During training, the critic can see the full portfolio state and learn whether the overall control decision is good. During execution, each group actor only uses its own building observations, plus any communication features, to choose continuous actions. After each episode, MAPPO uses the collected experience to estimate which actions were better than expected, and then updates the actors with PPO's conservative update rule.

The grouped replay buffer keeps the group-agent structure when samples are drawn for PPO updates. This matters for communication: the communication layer needs to know which agents were in the same group at the same time step. If the data is flattened too early, that structure is lost. The buffer therefore keeps tensors in a shape that supports both MAPPO updates and group communication.

3.6 Baseline and Communication Variants

The implemented methods are organized as a step-by-step experimental path. The project first tests individual learning methods, then standard MAPPO, then grouped MAPPO without communication, and finally several communication variants. This order makes it easier to see what is added at each stage.

3.6.1 Individual PPO and SAC Baselines

The first learning baselines are independent PPO and independent SAC. In these runs, each building learns without receiving information from other buildings. This gives a simple reference point for what can be achieved without multi-agent coordination.

3.6.2 Standard MAPPO

The next step is standard MAPPO. All 25 buildings reuse the same actor network through parameter sharing, while the centralized critic sees the full portfolio state during training. There is still no explicit communication module in this baseline. It tests whether a basic MARL method already improves the task before adding grouping or communication design.

3.6.3 Grouped MAPPO Without Communication

After standard MAPPO, buildings are divided into groups with K-means. Buildings in the same group share one group actor, but the encoded feature is not changed by any message passing, as shown in Equation (3.3):

$$\tilde{h}_i = h_i. \tag{3.3}$$

This tests whether grouping alone helps. It is also the direct no-communication baseline for later communication modules.

In the communication variants below, the actor first encodes each building observation into a hidden feature. The communication layer then modifies this feature before the action head selects the final action, as summarized in Equation (3.4):

$$o_i \to \text{encoder}_k \to h_i \to \text{communication} \to \tilde{h}_i \to \text{action head}_k \to a_i. \tag{3.4}$$

Because the communication module is part of the actor network, it is trained together with the actor.

3.6.4 CommNet-Style Mean Communication

For each group, the CommNet-style module lets each agent receive the average hidden feature of the other agents:

$$m_i = \frac{1}{|G_k| - 1} \sum_{j \in G_k, j \neq i} h_j. \tag{3.5}$$

This average message is passed through a small neural network and then added back to the agent's own feature:

$$\tilde{h}_i = h_i + f_\theta(m_i). \tag{3.6}$$

Equations (3.5) and (3.6) define the simplest communication baseline in the project.

3.6.5 Weighted Same-Group and Other-Group Communication

The project-specific communication module separates messages from similar buildings and messages from the rest of the portfolio. For group G_k,

$$m_k^{same} = \frac{1}{|G_k|} \sum_{j \in G_k} h_j, \qquad m_k^{other} = \frac{1}{N - |G_k|} \sum_{j \notin G_k} h_j. \tag{3.7}$$

The combined message is

$$m_k = \alpha m_k^{same} + \beta m_k^{other}, \qquad \alpha > \beta \geq 0. \tag{3.8}$$

The default implementation uses $\alpha = 0.75$ and $\beta = 0.25$. This means that same-group information is weighted more strongly, while other-group information is still kept as weaker global context, as defined in Equations (3.7) and (3.8).

3.6.6 PowerNet-Style Structured Information Passing

The PowerNet-style module tests a more structured way of sharing information. It can use ring, chain, or full topologies to decide which other agents are included in the message. For adjacency matrix A, the selected peer message is

$$m_i = \frac{1}{\max(1, \sum_j A_{ij})} \sum_j A_{ij} g_\theta(h_j). \tag{3.9}$$

The implementation then combines the agent's own feature with the peer message using a learned residual update. The purpose is to avoid overwriting local information while still allowing agents to coordinate, as captured in Equation (3.9).

3.6.7 GAT-Style Graph Attention Communication

The GAT-style module uses masked graph attention over a fixed similarity graph. Inside each K-means group, buildings are fully connected, and a small number of cross-group edges connect the closest groups. Each agent then learns attention weights over its allowed neighbors instead of using a simple average. This makes it possible to test whether graph-structured attention improves communication selectivity in the CityLearn setting.

3.6.8 TarMAC-Style Attention

The TarMAC-style module uses attention. Each agent produces query, key, and value features, and the attention weights are

$$\alpha_{ij} = \frac{\exp(q_i^\top k_j/\sqrt{d})}{\sum_l \exp(q_i^\top k_l/\sqrt{d})}, \tag{3.10}$$

and the received context is

$$c_i = \sum_j \alpha_{ij} v_j. \tag{3.11}$$

The updated feature is computed from h_i and c_i. Compared with the fixed α and β weights above, this module learns the communication weights from the current features. It tests whether the agents can learn which other agents are worth paying more attention to through Equations (3.10) and (3.11).

3.6.9 DIAL-Style Differentiable Communication

The DIAL-style module tests a different communication idea. Each agent first turns its hidden feature into a message. During training, the message remains continuous and differentiable, so gradients can pass through the communication channel. During evaluation, the message can be discretized. In this project, the DIAL module is adapted to MAPPO rather than implemented as the original DQN-based DIAL algorithm. The purpose is to test whether trainable communication signals help the grouped actors coordinate.

3.7 Completed Implementation

At the midterm stage, the following implementation work has been completed:

- RBC notebook baseline for non-learning comparison;
- Independent PPO and SAC training scripts;
- Standard MAPPO training and testing scripts, where all buildings reuse the same actor network through parameter sharing;
- Grouped MAPPO with K-means clustering and no communication;

- Intra-group communication variants, including no communication, CommNet-style mean communication, and PowerNet-style structured information passing;
- GAT-style graph attention communication over a fixed building-similarity graph;
- Weighted same-group and other-group communication, where same-group messages are weighted more strongly than messages from the rest of the portfolio;
- Attention and differentiable-message variants, including TarMAC-style attention and DIAL-style communication;
- Shared centralized critic and shared value normalizer for grouped policies;
- Grouped replay buffer preserving communication structure during PPO updates;
- Checkpoint loading and saving for test-only evaluation;
- Annex 96 metric export, daily metric tables, and plotting utilities, with cost and carbon fields marked as unavailable when the CE1 data provides no pricing or carbon-intensity signals.

The most important outcome so far is that the main methods can be run and compared through a common experimental interface. The completed Vermont benchmark results show that communication changes the balance between tracking, comfort, and secondary metrics. In the current selected benchmark set, `mappo_grouped_powernet_global` is the strongest overall method. The ranking tables also show a more nuanced picture: the default weighted communication variant is strongest on the primary-only ranking, while TarMAC and weighted communication with $\alpha = 0.90, \beta = 0.10$ remain competitive all-round options.

3.8 Current Limitations

Although the current method variants already show clear progress beyond the basic baselines, this part of the project is still not finished. A useful reference point is the independent PPO baseline. Its advantage is that it is simple, stable, and easy to train and debug, and in the current results it still remains competitive on some metrics. Its weakness is that each building learns largely on its own, so it cannot model portfolio-level coordination directly. In addition, PPO has a high training-time cost in multi-agent settings, and this usually becomes more severe as the number of agents grows; in comparison with shared-structure methods such as MAPPO, this is a major disadvantage. The current grouped MAPPO variants show that moving beyond single PPO is worthwhile, especially when the goal is to improve overall coordination and to study communication design, but the comparison also shows that independent PPO is not a trivial baseline and has not been dominated in every metric.

Another limitation comes from the current K-means grouping and the TarMAC-style attention mechanism. In the current implementation, K-means determines not only the building groups but also the actor-sharing structure, while TarMAC only learns dynamic attention weights within this fixed grouped-actor framework. These two ideas are not fully aligned: hard grouping can simplify training and control parameter size, but it may also restrict the attention module from discovering more useful cross-group relationships. Therefore, the current pipeline still does not answer which grouping structure is most suitable for attention-based communication, or whether softer and learnable actor-assignment mechanisms should be studied in the future.

There are also several model variants that have not yet been tested. One obvious next step is a hybrid of TarMAC and PowerNet, so that the model can combine structured message passing with learned attention instead of treating them as separate alternatives. Another open direction is PPO-structure variation. The current implementation mainly uses one centralized critic, but variants such as multiple critics, group-specific critics, or mixed local-global critics may be better suited to grouped communication settings, because multi-critic designs may lead to more stable and more conservative parameter updates. These architectural questions are still open [11, 12].

Finally, there are still some extension studies that have not yet been fully completed. In particular, Texas RL training with the flatter district target has not yet been included in the main comparison, and multi-seed validation is also still pending.

Chapter 4

Evaluation and Result Analysis

This chapter reports the Vermont benchmark comparison based on the selected experiment set. The numerical statements below are based on the consolidated outputs from `summarize_selected_experiment_metrics`, together with the exported `results`, `wandb`, and `target_folder` files.

4.1 Evaluation Scope

The experiments use the Annex 96 Common Exercise 1 CityLearn portfolios. Each portfolio contains 25 single-family homes with photovoltaic generation and battery storage. The fully summarized comparison in this report is the Vermont setting, where January is used for training and February is used for testing. Texas remains future work for the RL comparison, so the conclusions in this chapter should be read as Vermont benchmark conclusions rather than cross-climate conclusions. Table 4.1 summarizes the current train and test split used in this report.

Climate	Training month	Testing month	Current status
Vermont	January	February	fully summarized in this report
Texas	August	September	RL comparison remains future work

Table 4.1: Annex 96 train and test split and current evaluation scope.

The selected Vermont comparison contains 15 representative runs: RBC, independent PPO and SAC baselines, standard MAPPO, grouped MAPPO, and the main communication variants. Under the current hardware constraints, most MAPPO experiments in the selected set were trained for about 500 episodes and reached rough convergence, although some communication methods such as TarMAC may still improve further. The independent PPO and SAC results are also useful references, but because these methods require more training time and higher computational cost, in this report they are mainly used as important references after the sum reward and each agent's reward had roughly converged.

4.2 Overall Ranking Summary

The ranking tables show that there is currently no method that is absolutely best under every reading of the task. In the current Vermont result set, `mappo_grouped_powernet_global` ranks first in the overall ranking. The secondary metrics are not explicitly included in the reward function, but they are still useful as references, because they show which methods already behave well on those objectives even without being directly optimized. However, if the comparison is restricted to the primary metrics, namely CV-RMSE, |NMBE|, and comfort, the top method becomes `mappo_grouped_comm_weighted_default`. Table 4.2 reports the primary-only ranking.

Controller	Primary rank	Primary score	CV-RMSE (%)	|NMBE| (%)	Comfort hours
Grouped MAPPO + Weighted Default	1	3.33	48.68	2.31	3166
Grouped MAPPO + PowerNet Global	2	3.75	46.45	0.43	4359
RLlib Independent PPO	3	4.00	49.76	1.24	2309
Grouped MAPPO + Weighted $(0.90, 0.10)$	4	4.83	49.75	1.95	3308
Grouped MAPPO + Weighted $(0.55, 0.45)$	5	5.33	49.28	0.27	5376
Grouped MAPPO	6	6.92	51.50	2.75	3520
Grouped MAPPO + TarMAC	7	7.00	49.65	3.32	3713
Grouped MAPPO + GAT	8	7.75	51.35	2.92	3866
Grouped MAPPO + CommNet	9	8.75	48.94	8.11	4873
RLlib SAC	10	9.25	46.53	8.93	7470
Grouped MAPPO + Comm v2	11	10.08	52.92	3.80	4031
Standard MAPPO	12	10.42	53.30	5.29	3542
Grouped MAPPO + DIAL	13	11.33	52.39	7.25	6755
Grouped MAPPO + PowerNet	14	12.42	50.16	9.15	7685
RBC baseline	15	14.83	138.27	109.37	11465

Table 4.2: Primary-only ranking for the selected Vermont experiments.

The primary-only ranking gives a clearer picture of the main tracking objective. The best primary-score balance now comes from the default weighted communication setting, here the default weights are $\alpha = 0.75, \beta = 0.25$. It combines low CV-RMSE, moderate bias, and a strong comfort result. The PowerNet Global variant ranks second because it has the best CV-RMSE and the smallest absolute NMBE in the main MAPPO set, but it gives up more comfort than the weighted default controller. The independent PPO baseline ranks third and has especially strong comfort performance, but its CV-RMSE and absolute NMBE are weaker than the top two methods, likely because it does not model explicit inter-building coordination and therefore cannot optimize the portfolio-level tracking objective as directly as the grouped communication methods.

For completeness, Table 4.3 reports the overall composite ranking generated by `summarize_selected_experiment_metrics`. Unlike the primary-only table above, this ranking also reflects selected secondary objectives, so it is useful for understanding why the final "overall best" method is not exactly the same as the primary-only winner.

Controller	Overall rank	Overall score	Primary rank	Recommended rank	Peak change (%)	Ramping (kW)
Grouped MAPPO + PowerNet Global	1	5.14	2	3	-4.19	478.22
RLlib SAC	2	5.60	10	7	-18.80	353.48
RLlib Independent PPO	3	5.87	3	1	-2.69	372.06
Grouped MAPPO + TarMAC	4	6.22	7	6	-1.38	486.31
Grouped MAPPO + Weighted Default	5	6.48	1	2	5.13	429.05
Grouped MAPPO + CommNet	6	6.77	9	8	-4.45	441.78
Grouped MAPPO + Weighted $(0.90, 0.10)$	7	6.98	4	4	-2.20	443.12
Grouped MAPPO + Weighted $(0.55, 0.45)$	8	8.15	5	5	0.54	463.45
Grouped MAPPO	9	8.19	6	9	-1.19	519.14
Grouped MAPPO + Comm v2	10	9.03	11	12	-0.05	498.03
Grouped MAPPO + DIAL	11	9.19	13	11	-2.04	498.95
Grouped MAPPO + GAT	12	9.29	8	10	-0.15	537.38
Grouped MAPPO + PowerNet	13	9.64	14	14	-0.63	459.18
Standard MAPPO	14	10.40	12	13	9.28	506.42
RBC baseline	15	13.06	15	15	-8.44	–

Table 4.3: Overall composite ranking for the selected Vermont experiments.

4.3 Primary and Secondary Metric Overview

The primary objective is portfolio-level reference tracking. Let y_t be the actual aggregate portfolio load and r_t be the reference load. The two main tracking metrics are NMBE and CV-RMSE, and their definitions are given below:

$$\text{NMBE} = \frac{\frac{1}{T}\sum_{t=1}^{T}(y_t - r_t)}{\frac{1}{T}\sum_{t=1}^{T} r_t} \times 100\%. \tag{4.1}$$

$$\text{CV-RMSE} = \frac{\sqrt{\frac{1}{T}\sum_{t=1}^{T}(y_t - r_t)^2}}{\frac{1}{T}\sum_{t=1}^{T} r_t} \times 100\%. \tag{4.2}$$

Lower absolute NMBE and lower CV-RMSE in Equations (4.1) and (4.2) mean better tracking. At the same time, comfort exceedance hours, here mainly temperatures above 24°C or below 20°C, must also be reported, because a controller that tracks the target well but causes many temperature violations is still not a good practical solution.

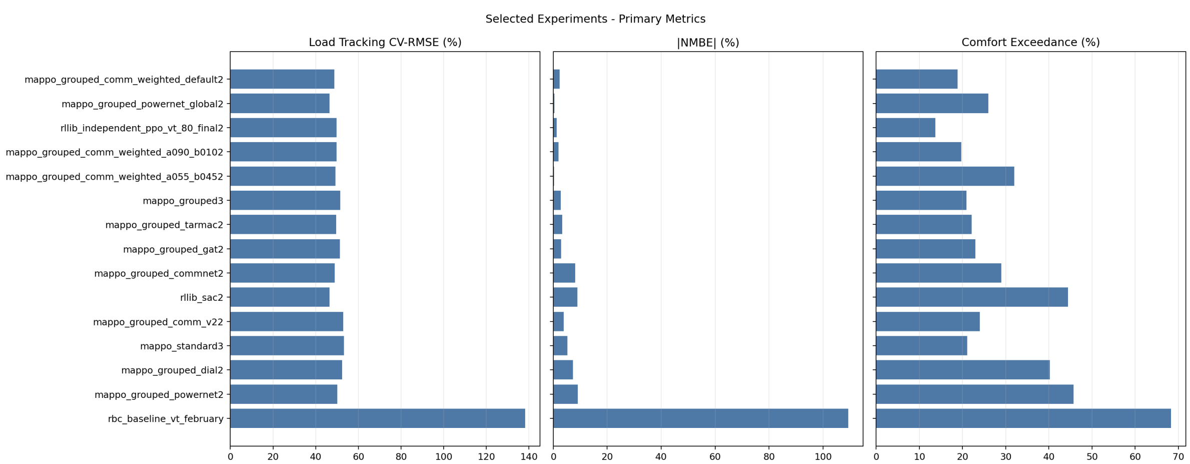

Figure 4.1: Primary metric summary for the selected Vermont experiments.

Figures 4.1 and 4.2 show the main trade-offs in the current results very clearly. The lowest CV-RMSE now comes from PowerNet Global and SAC, but SAC is clearly worse in comfort and NMBE. The weighted default communication run is not first overall, but it is the most balanced method on the primary metrics because it combines a low CV-RMSE with the fewest comfort-exceedance hours among the main MAPPO communication methods. PowerNet Global ranks first overall because, even without secondary metrics being trained explicitly, it has no obvious weakness across several categories: it is second on the primary table, fourth on peak reduction, and sixth on fairness.

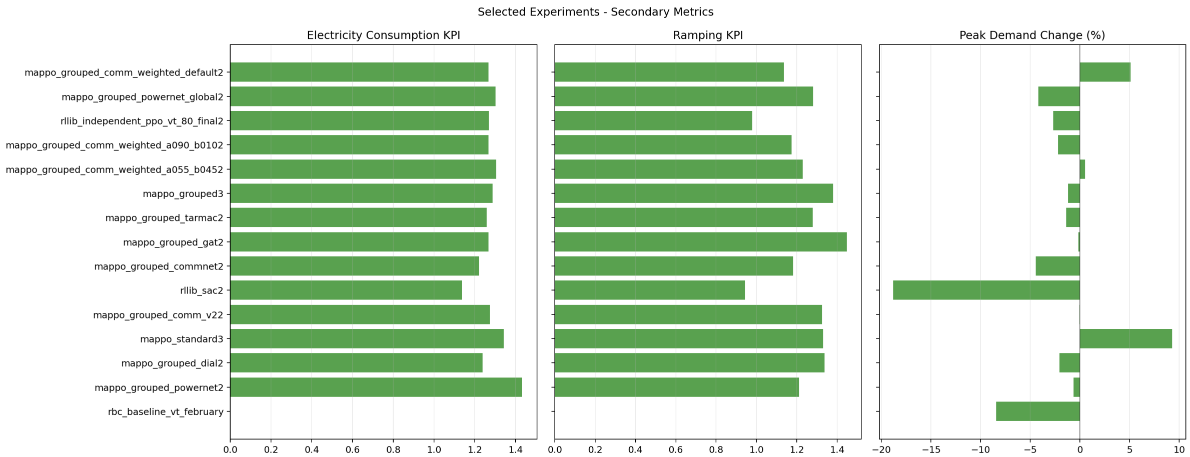

Figure 4.2: Secondary metric summary for the selected Vermont experiments.

4.4 Best Run Diagnostics

Because the overall best controller in this result set is `mappo_grouped_powernet_global`, it is useful to look at its exported plots directly.

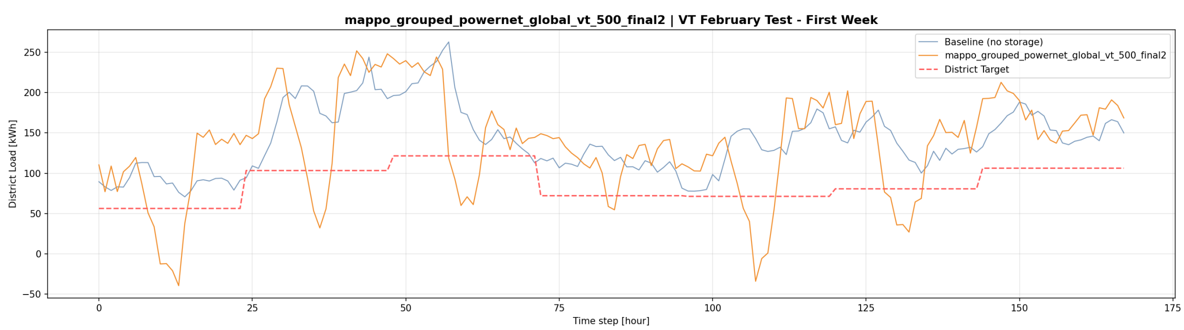

Figure 4.3: Week-1 aggregate load tracking for the best overall Vermont run.

Figures 4.3 and 4.4 show why this controller is strong but not perfect. It reacts more aggressively than the no-storage baseline and is able to shift the aggregate profile away from some high-load periods, which helps the final CV-RMSE and peak-demand reduction. At the same time, the response is sometimes too aggressive: the full-month plot contains several deep downward excursions, including negative net-load periods. This is consistent with the secondary metrics, where the same run is good on peak demand but weak on ramping and peak-to-valley

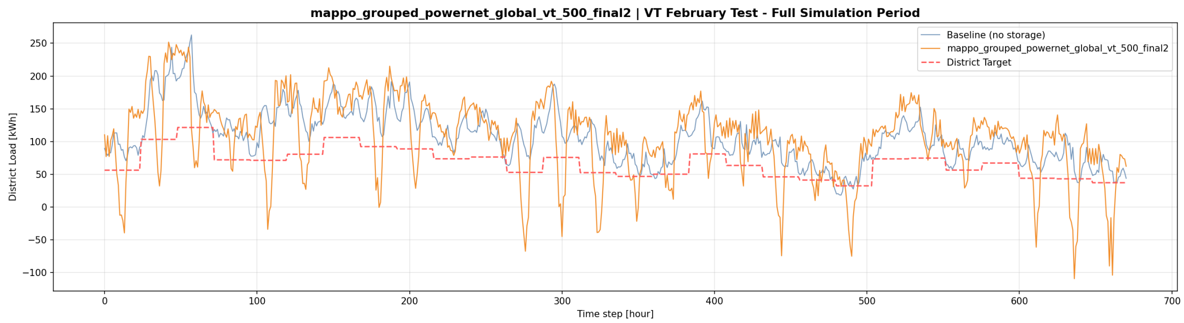

Figure 4.4: Full-month aggregate load tracking for the best overall Vermont run.

ratio.

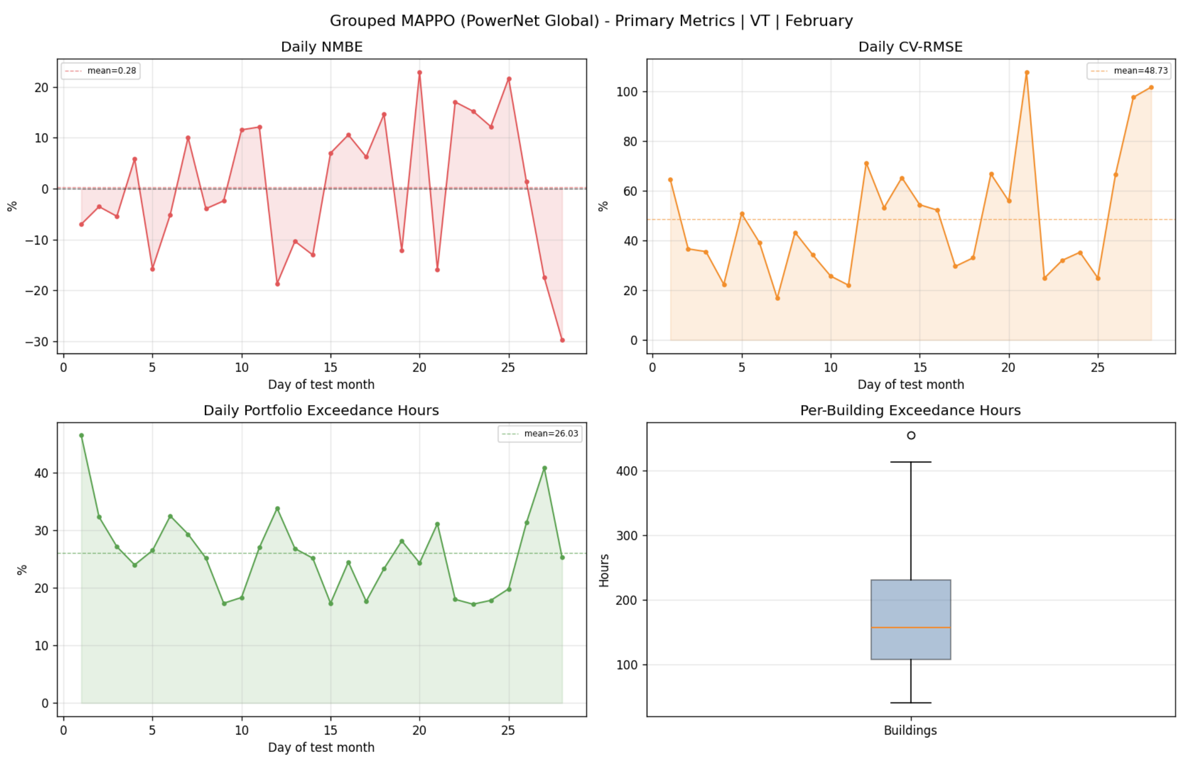

Figure 4.5: Daily primary metrics for the best overall Vermont run.

Figure 4.5 gives the same conclusion in a more stable form. For the best overall run, the daily NMBE mean is already close to zero, the daily CV-RMSE mean is about 48.7%, and the daily portfolio comfort exceedance mean is about 26.0%. However, the day-to-day variation is still quite large, and some days are clearly better than others. The per-building comfort boxplot also shows a wide spread: a few buildings contribute a large share of the total violation hours. In this run, the worst building reaches about 454 exceedance hours, compared with a mean of about 174.9 hours, which is roughly 2.6 times the average.

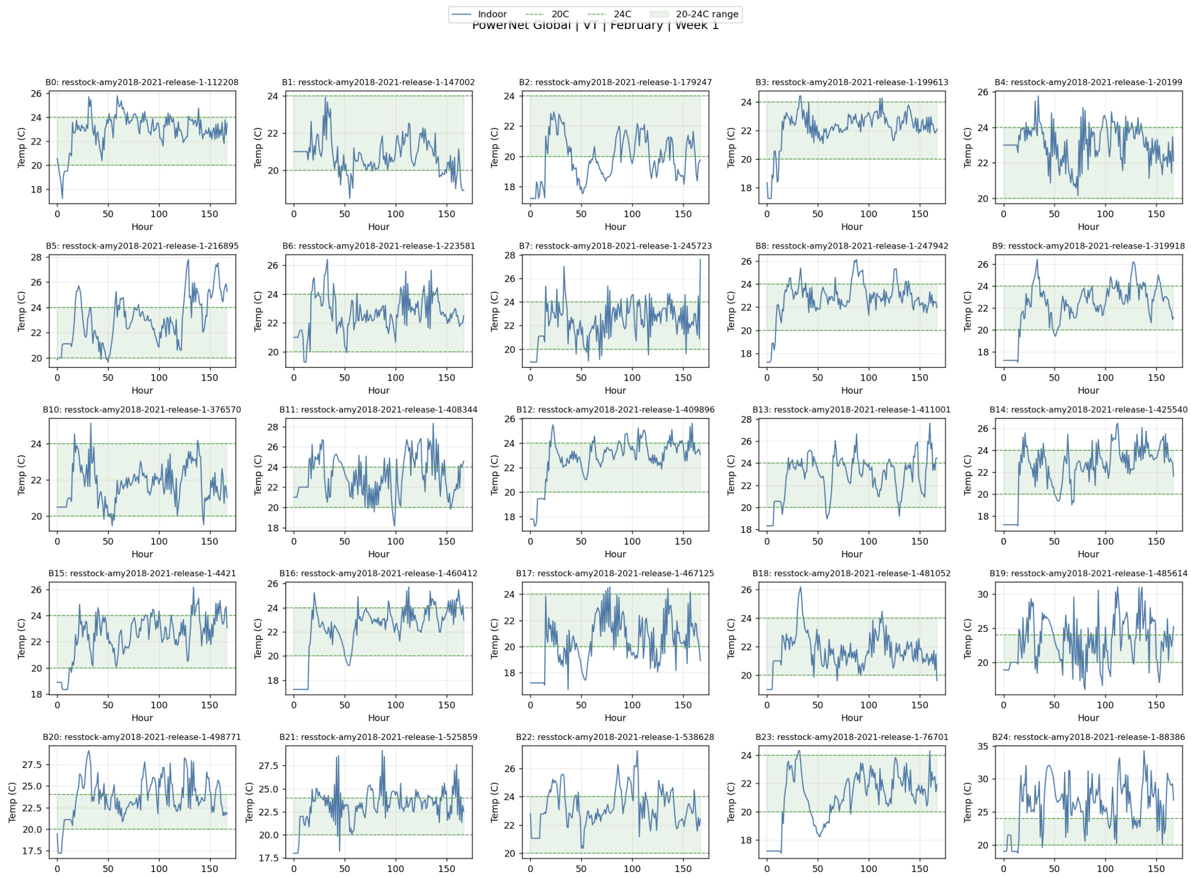

Figure 4.6: Week-1 building temperature traces for the best overall Vermont run.

Figures 4.6 and 4.7 show that comfort is still the hardest part of the task. Several homes stay near the comfort band (20C-24C) for most of the month, but some homes spend long periods above the upper bound. This is exactly why the best overall controller is not the best comfort controller. In the exported comfort table, the worst individual buildings still exceed 400 hours, while the easiest buildings stay below 100 hours.

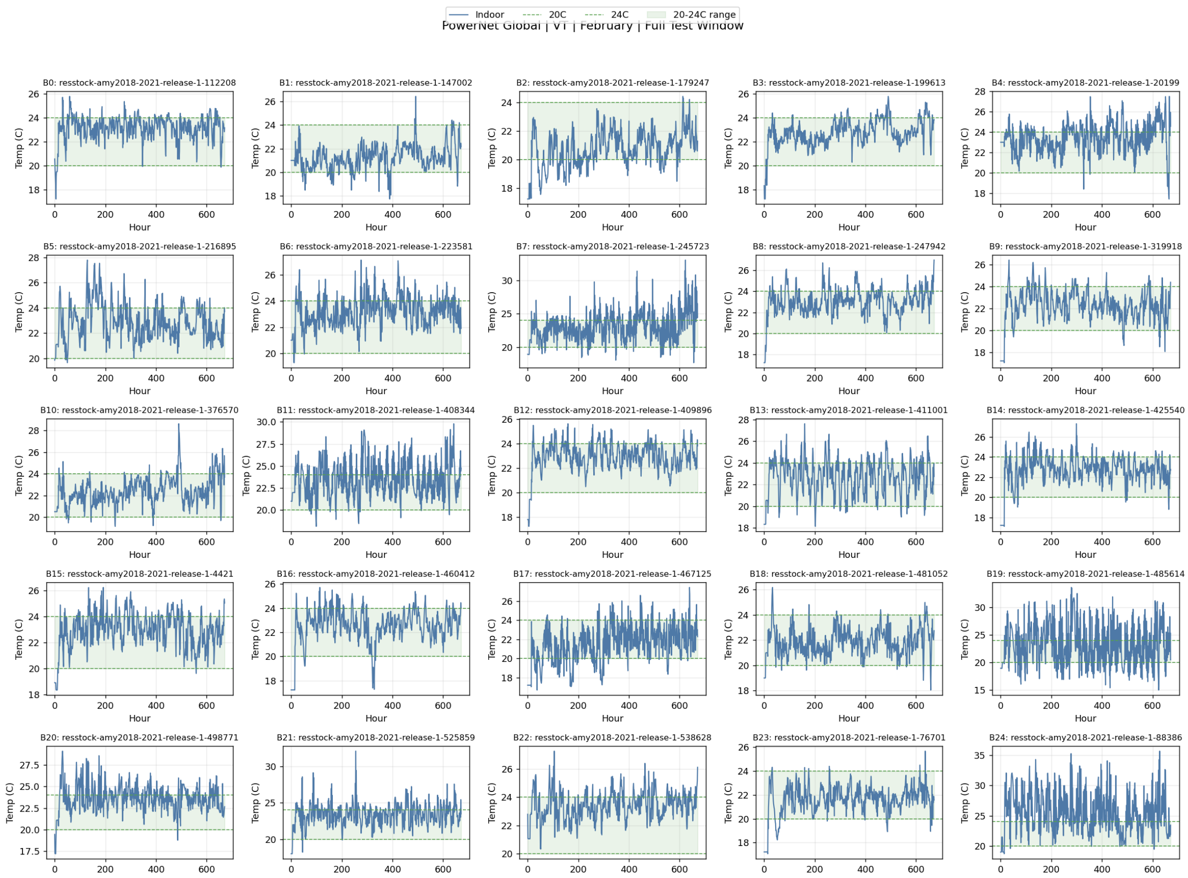

Figure 4.7: Full-month building temperature traces for the best overall Vermont run.

4.5 Communication Variant Comparison

Table 4.4 focuses only on the grouped communication variants.

Method	Overall	Primary	Recommended	CV-RMSE (%)	Comfort hours	Peak change (%)
PowerNet Global	1	2	3	46.45	4359	-4.19
TarMAC	4	7	6	49.65	3713	-1.38
Weighted default	5	1	2	48.68	3166	5.13
CommNet	6	9	8	48.94	4873	-4.45
Weighted $(0.90, 0.10)$	7	4	4	49.75	3308	-2.20
Weighted $(0.55, 0.45)$	8	5	5	49.28	5376	0.54
Comm v2	10	11	12	52.92	4031	-0.05
DIAL	11	13	11	52.39	6755	-2.04
GAT	12	8	10	51.35	3866	-0.15
PowerNet	13	14	14	50.16	7685	-0.63

Table 4.4: Vermont comparison for grouped communication variants.

Based on the current results, several communication conclusions can be stated more clearly. First, the strongest overall communication variant is PowerNet Global. One direct interpretation is that when the PowerNet-style block adds structured information from other agents, it first retains local features; this can be understood as a form of information filtering or cropping, so useful information is less likely to be overwritten during communication [3]. Second, the weighted communication family shows that the weight setting itself has a clear effect on the final result. Information from other groups can sometimes act as noise, but it can also contain important portfolio-level information, so changing the same-group and other-group weights changes the result substantially. Third, TarMAC can be seen as an upgraded version of weighted communication: instead of using fixed weights, it uses attention to let each agent learn which other buildings matter more in the current state [5]. However, in the current implementation, the underlying communication in TarMAC is still closer to CommNet-style mean aggregation and does not yet incorporate the more selective PowerNet-style structured information filtering or cropping [17]. This is why combining TarMAC with PowerNet remains a promising direction. Finally, DIAL was originally proposed mainly for DQN-style discrete-action settings. Its core idea is that differentiable messages let agents see the feedback produced by their own communication choices; during training, continuous messages are used to pass gradients, and at execution time a sigmoid-like mechanism is used to approximate discrete messages [7]. In the current MAPPO-based continuous-control setting, this adapted DIAL performs poorly, which suggests that the original DIAL idea does not transfer directly here.

This communication table also shows why it is necessary to distinguish "overall best," "best comfort," and "best primary score." In the current setting, the secondary metrics are not explicitly included in the reward function. The weighted default controller gives the best primary score and the fewest comfort exceedance hours in Table 4.4, but it also pushes peak demand above the baseline. This means that the secondary-metric differences seen here are more natural results brought out by different communication structures. PowerNet Global is not the best in comfort, but it is more balanced across the primary objective and some secondary metrics, so it ultimately ranks first overall. This also gives a useful direction for future work that may explicitly include secondary metrics in the training objective.

4.6 Weighted Communication Ablation

The weighted communication family is still valuable because it studies several very specific questions on their own: how same-group and other-group messages should be weighted, and whether it is necessary to let the agents learn those weights themselves through attention. Table 4.5 summarizes the corresponding ablation results.

Weighted setting	Overall	Primary	CV-RMSE (%)	Comfort hours	Peak change (%)
Default $(0.75, 0.25)$	5	1	48.68	3166	5.13
$(0.90, 0.10)$	7	4	49.75	3308	-2.20
$(0.55, 0.45)$	8	5	49.28	5376	0.54

Table 4.5: Updated weighted communication ablation for Vermont.

This ablation shows that the simple claim "putting more weight on same-group information is always better" is wrong. The exported results suggest that only under specific weights does communication work best, and only then does information from other groups stop acting like noise and start acting like a gain. The default-weight setting $\alpha = 0.75, \beta = 0.25$ is the strongest on the primary metrics and also has the fewest comfort exceedance hours. The $(0.90, 0.10)$

setting is slightly weaker on the pure primary metrics, but it is more balanced. The more even $(0.55, 0.45)$ setting almost removes NMBE bias, but its comfort result becomes clearly worse.

4.7 Rule-Based Baseline

Table 4.6 reports the available RBC notebook comparison for Texas and Vermont. The RBC baseline is still useful as a non-learning reference, but it is clearly much weaker than the learned controllers on the CE1 load-tracking task.

Climate	NMBE RBC (%)	NMBE baseline (%)	CV-RMSE RBC (%)	CV-RMSE baseline (%)
TX	89.57	79.12	207.63	163.74
VT	103.93	101.11	129.09	144.56

Table 4.6: RBC notebook baseline comparison against the CE1 target profile.

The RBC results still support the same broad conclusion: fixed hour-based charge/discharge rules are not enough for the CE1 objective. In Vermont, RBC improves CV-RMSE relative to the no-control baseline, but its bias remains extremely large. In Texas, it is worse than the baseline on both NMBE and CV-RMSE.

4.8 Current Interpretation and Remaining Limits

The Vermont results support four main conclusions. First, MARL is helpful for this task. The stronger grouped MAPPO variants clearly outperform RBC, and compared with independently trained PPO and SAC they also better show the advantage of portfolio-level cooperative control; in particular, they avoid the larger comfort cost seen in SAC and also make up for the lack of explicit coordination in independent PPO. Second, communication clearly matters, but there is no fixed answer to which communication method is best. In the current results, PowerNet Global is the strongest all-round communication variant. This suggests that, when structured shared information is added, applying more selective filtering or cropping to that shared information is an effective direction. Third, the weighted communication family shows that message weights themselves clearly affect the result: information from other groups can either act as noise or contain important portfolio-level context. Fourth, the results of TarMAC and DIAL show that not every trainable communication mechanism transfers equally smoothly to this setting. Attention-based weighting remains promising, especially after it is further combined with PowerNet-style information filtering, whereas the current MAPPO-based adaptation of DIAL performs poorly and suggests that the original DIAL idea cannot simply be copied into this continuous-control setting.

The main remaining limitation is that the communication methods still need to be upgraded and studied in more integrated forms. Based on that, future work should carry out more targeted and more sufficient training together with reward-function parameter tuning, especially by combining attention-based weighting with more selective PowerNet-style information filtering. It should also test stronger critic-structure variants to obtain more stable parameter updates, and extend the comparison to other MARL families such as MASAC or value-based methods.

Chapter 5

Conclusion and Work Plan

This midterm report has focused on one central question: in the CityLearn portfolio load-tracking task, what kinds of MARL structures and communication designs are more effective, and why. The earlier chapters reviewed the relevant CityLearn and MARL literature, explained the CE1 benchmark setting, and described the implemented grouped-control and communication pipeline.

The implementation work is now sufficient to support a meaningful midterm conclusion. The codebase includes RBC, independent PPO/SAC, standard MAPPO, grouped MAPPO, K-means-based actor grouping, and several communication-aware variants such as CommNet, weighted communication, PowerNet, TarMAC, DIAL, and GAT. The evaluation pipeline exports Annex 96 metrics, ranking tables, and the main diagnostic plots used in this report. The selected Vermont benchmark set has also been completed. In the current report, the reward function is designed around the three primary objectives, namely CV-RMSE, |NMBE|, and comfort, while the secondary metrics are used only for evaluation and comparison rather than direct optimization.

The main conclusion at this stage is that structural choices already affect the result substantially. First, MARL is useful for this task: the stronger grouped MAPPO variants clearly outperform RBC and show portfolio-level coordination more directly than independently trained controllers. Second, communication is not a detail but a core design variable. Different communication structures change the trade-off between tracking accuracy, comfort, and secondary metrics in different ways, so the final conclusion depends on which objective mix is emphasized. In the current Vermont results, `mappo_grouped_comm_weighted_default` is the strongest method on the primary-only ranking, and the weighted communication family shows that whether information from other groups acts as a gain or as noise depends heavily on the communication weights. Further, `mappo_grouped_powernet_global` gives the strongest overall result, which supports the idea that more selective filtering or cropping of shared information is useful. Third, many communication methods still have large room for improvement. For example, TarMAC remains promising as a learnable weighting mechanism, but its current implementation has not yet been combined with the PowerNet-style filtering idea. In addition, more critic structure combinations should be tested to compare their effects.

5.1 Next Steps

The work plan for the next stage is:

- Extend the current comparison from Vermont to Texas; the Texas training pipeline has already been implemented, and the main remaining step is to run and summarize those experiments under the same evaluation framework;
- Add multi-seed evaluation for the strongest methods, especially standard MAPPO, grouped MAPPO, PowerNet Global, TarMAC, weighted communication, and CommNet, so that the current conclusions are supported by variance-aware comparisons rather than single runs;
- Continue the communication study itself, especially by combining attention-based weighting with PowerNet-style selective information filtering, and by comparing fixed weighting, learned weighting, and more integrated communication designs;
- Test stronger critic-structure and PPO-structure variants, and compare the current MAPPO backbone with other MARL families such as MASAC and value-based methods;
- Analyse aggregate load tracking, comfort, ramping, battery behaviour, and fairness in more detail;

- Refine the final report with the completed Texas results, additional sensitivity tests, and a clearer explanation of why the stronger communication designs work better.

If time allows, additional work will test sensitivity to the number of clusters, communication rounds, and weighted communication coefficients. There is also the idea of replacing the lower-level MLP with an LSTM. Since LSTM is a structure better suited to time-series data, it may further improve performance [10]. A further extension is to learn building embeddings for grouping instead of relying only on static flexibility features, so that grouping itself may become more adaptive to the control task and may help address the limitation that static K-means grouping can hold back TarMAC-style attention.

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